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REVIEW ARTICLE

Emerging Targeted Therapies for Early Breast Cancer

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Abstract

Breast cancer is the most common malignancy and the second leading cause of cancer-related mortality in the United States (US). Most patients are diagnosed with early-stage disease; however, there is still a need to prevent recurrences that often present as incurable metastatic disease. The treatment landscape of early-stage breast cancer is evolving rapidly. The immune checkpoint inhibitor pembrolizumab is approved in combination with neoadjuvant chemotherapy for the treatment of high-risk triple-negative breast cancer (TNBC). The cyclin-dependent kinase (CDK) 4 and 6 inhibitor abemaciclib is approved for adjuvant treatment of patients with high-risk hormone receptor (HR)-positive disease. While adjuvant olaparib has shown significant improvement in outcomes for patients with pathogenic/likely pathogenic *BRCA1/2* mutations and high-risk human epidermal growth factor receptor 2 (HER2)-negative breast cancer, and is approved in this setting. For the HER2-positive subtype, the post-neoadjuvant therapy can be tailored based on the response to neoadjuvant chemotherapy and HER2-targeted agents. In this narrative review, we summarize the most recent approvals for early-stage breast cancer as well as frequently encountered clinical challenges utilizing these medications.

Key Points

The treatment options for patients with early breast cancer are evolving rapidly, however new strategies are needed to prevent cancer recurrences that often present as metastatic disease.

Preoperative immunotherapy in combination with chemotherapy is the current standard of care for many patients with early triple-negative breast cancer, however many questions remain to be answered in order to select patients who can benefit from this treatment and preventing long-term toxicities.

Individualizing treatment based not only on breast cancer subtype but also on response to therapy and clinicopathologic features may help improve patients while minimizing risk for toxicities for those with low-risk disease.

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1 Introduction

Breast cancer is the most common malignancy and the second leading cause of cancer-related mortality in the United States (US) [1]. Over 90% of patients with breast cancer in US are diagnosed with early disease, including 61% with disease localized to the breast and 32% with breast and local lymph node involvement [2, 3]. The prognosis of these patients has improved over time, as a result of earlier detection and improved cancer treatments [2, 4]; however, there is still a need to prevent recurrences that often present as incurable metastatic disease [5–7].

Systemic treatment for early breast cancer can be administered before definitive surgery (neoadjuvant) or after surgery (adjuvant therapy). Several studies from the 1980s and 1990s revealed that breast cancer disease-free survival (DFS) and overall survival (OS) outcomes were similar for patients who received neoadjuvant or adjuvant therapies. Of note, in the mentioned studies, patients received the same regimens in the (neo)adjuvant settings and the treatment was not modified based on response to therapy [8, 9]. A meta-analysis by the Early Breast Cancer Trialist Collaborative Group (EBCTCG) included 4756 women randomized to receive neoadjuvant or adjuvant chemotherapy between 1983 and 2002 [9]. This analysis confirmed no differences in distant recurrence rates or

breast cancer mortality between the therapy groups [9]. Historically, the decision about the timing of systemic therapy was based on patient and physician preferences and to optimize surgical and cosmetic outcomes. However, over the past few years, there has been a shift towards neoadjuvant systemic therapy, particularly in the triple-negative and human epidermal growth factor receptor 2 (HER2)-positive breast cancer subtypes, which have demonstrated sensitivity to neoadjuvant chemotherapy. Several advantages to neoadjuvant systemic therapy have supported this shift, including the potential increase in the conversion rate from mastectomy to breast-conserving surgery, the in vivo assessment of tumor sensitivity to anti-cancer drugs, and prognostic risk stratification. Indeed, pathologic complete response (pCR), i.e. the complete disappearance of invasive cancer in the breast and/or regional lymph nodes at the time of surgery, has increasingly been used as a surrogate of long-term outcomes at the patient level [10], allowing treatment intensification based on a risk-adapted approach in the post-neoadjuvant setting, based on the response to neoadjuvant treatment [5, 6, 11, 12]. Several novel systemic therapies have been approved to treat early-stage breast cancer over the past few years. In this review, we discuss the recent approvals, limitations, and current use in clinical practice.

2 Triple-Negative Breast Cancer (TNBC)

Triple-negative breast cancer (TNBC) is characterized by the absence of estrogen and progesterone receptors (hormone receptors [HRs]) and HER2 overexpression, and accounts for 15–20% of all breast cancers [13]. TNBC has been associated with a worse prognosis relative to other breast cancer subtypes [13]. The systemic treatment for early-stage TNBC has for decades been limited to cytotoxic chemotherapy [14], most commonly including anthracyclines, cyclophosphamide, and taxanes, with or without the addition of platinum salts. However, in July 2021, the immune checkpoint inhibitor (ICI) anti-programmed cell death 1 (PD-1) monoclonal antibody (mAb) pembrolizumab was approved by the US FDA in combination with chemotherapy, based on the results from the KEYNOTE-522 trial [5, 12], reshaping treatment for early TNBC. In this phase III study, 1174 patients with stage II–III (T1c, N1-2 or T2-4, N0-2) TNBC were randomized to receive combination chemotherapy with or without pembrolizumab [5, 12]. Patients who received pembrolizumab in the neoadjuvant setting then received 6 months of post-neoadjuvant pembrolizumab monotherapy irrespective of pCR [5, 12]. With a median follow-up of 39 months, the trial showed an improvement in the 3-year event-free survival (EFS), favoring pembrolizumab (84.5 vs. 76.8%; hazard ratio [HR] 0.63, 95% confidence interval [CI] 0.48–0.82, $p = 0.00031$); 3-year distant recurrence-free survival (RFS) was also better in the ICI arm (87.0 vs. 80.7%; HR 0.61, 95% CI 0.46–0.82). The OS was not significantly improved, although a numerical trend was reported favoring pembrolizumab (89.7 vs. 86.9%; HR 0.72, 95% CI 0.51–1.02, $p = 0.032$), and longer follow-up is needed [5, 12]. The addition of pembrolizumab also improved pCR rates, with an absolute gain of 13.6% (64.8 vs. 51.2%); an unpublished updated analysis conducted on the overall population showed a lower 7% difference (63 vs. 56%), still statistically significant [15, 16]. Similar to what has been observed with neoadjuvant chemotherapy, patients with residual disease at surgery had a higher rate of recurrence when compared with those who achieved a pCR [5, 12]. Notably, the benefit of pembrolizumab was demonstrated irrespective of programmed cell death ligand 1 (PD-L1) expression (PD-L1 positivity was defined as a combined positive score of ≥ 1 by the 22C3 assay) or nodal status [5, 12]. A recent analysis of EFS by residual cancer burden (RCB) after neoadjuvant chemoimmunotherapy revealed that patients with higher disease burden derived a greater benefit from the addition of pembrolizumab [16]. A neoadjuvant regimen that combines paclitaxel, carboplatin, an anthracycline (doxorubicin or epirubicin), and cyclophosphamide with pembrolizumab followed by adjuvant therapy with pembrolizumab is now considered the standard of care in the treatment of most patients with tumors larger than 2 cm or with any lymph node involvement. Patients with smaller and node-negative tumors are often treated with chemotherapy alone [14]. Although the KEYNOTE-522 regimen has changed the treatment paradigm of early-stage TNBC, several questions still have to be answered.

2.1 Patient Selection

The chemoimmunotherapy regimen studied in KEYNOTE-522 trial is the current standard of care for the treatment of patients with high-risk TNBC, and factors such as tumor size, nodal status and PD-L1 expression were not predictive of response in this trial [5, 12]. Given that there is a risk for significant and potentially long-term toxicities, there is an urgent need to identify predictive biomarkers to determine which patients are more likely to benefit from the addition to chemotherapy [5, 12].

PD-L1 expression has been studied broadly as a biomarker of response to ICIs in multiple malignancies and was predictive of response to immunotherapy in metastatic TNBC [17, 18]; therefore, ICIs have only been approved for that subset of patients with advanced disease. Conversely,

in patients with early-stage disease, the predictive role of PD-L1 is less clear. While there was no apparent benefit in the IMpassion031 or KEYNOTE studies, there was a signal in the NeoTRIPDL1 and KEYNOTE-173 trials [12, 19–21]. In KEYNOTE 522, there was a numerical increase in pCR rates with increased PD-L1 expression in both treatment arms (with or without pembrolizumab), suggesting that PD-L1 is predictive of response to neoadjuvant therapy, not specifically for ICI benefit [18]. Another factor that needs to be considered is the assay selected to quantify PD-L1 expression. Three assays are available for PD-L1 (VENTANA SP142, SP263, Dako 22C3) and it has been shown that the results are not always concordant; therefore, when possible [22], the assay used in the clinical trial should be based on the ICI being used. PD-L1 expression is not currently being used to guide therapeutic decisions in early-stage disease.

Other potential predictive biomarkers have been studied. The presence of high quantities of tumor-infiltrating lymphocytes (TILs) has shown to be predictive of response to neoadjuvant therapy [23]. An analysis of the phase I KEYNOTE-173 study showed that higher quantities of stromal TILs were associated with greater pCR rates [21]. However, in this study, all patients received neoadjuvant pembrolizumab and chemotherapy, and TILs have also shown to be predictive of response to chemotherapy alone [24]. Additional information is needed to determine if TILs could be used to identify patients who may benefit from the addition of ICIs to chemotherapy.

The TNBC subtypes [25] have also been studied as biomarkers of response to ICIs. An analysis of the IMpassion130 study showed that only the basal-like immune-activated (BLIA) and immune-suppressed (BLIS) TNBC subtypes derived a benefit when an ICI was added to chemotherapy in patients with metastatic TNBC [26]. There are known differences between primary and metastatic tumors in PD-L1 expression and in the tumor immune microenvironment (TME), therefore one has to be careful extrapolating data from advanced disease when selecting neoadjuvant therapies [27, 28]. Additional analysis of previously published and ongoing studies will likely help elucidate which patients can benefit from the addition of an ICI to their therapy.

In terms of toxicities, in KEYNOTE-522, non-immune-mediated toxicities were similar between the experimental and control arms; however, there was a higher percentage of toxicity-related treatment discontinuations in the pembrolizumab group (14.1 vs 27.7%) [5]. Immune-mediated adverse events (AEs) were more common in the pembrolizumab arm, particularly infusion-related reactions (11.6 vs. 18%), hypothyroidism (5.7 vs. 15.1%), severe skin reactions (1 vs. 5.8%), hyperthyroidism (1.8 vs. 5.2%), and adrenal insufficiency (0 vs. 2.6%) [5]. These differences were more significant in the neoadjuvant phase of the study [5]. Notably, endocrine immune toxicities have the potential to be irreversible, and the incidence of these complications appears to be higher in patients with breast cancer [5, 17] when compared with other solid tumors, such as lung cancer or melanoma treated with ICIs [29, 30]; the reasons for these differences are yet to be elucidated.

2.2 Neoadjuvant vs. Adjuvant Immunotherapy

According to preclinical models, the release of tumor-specific antigens and subsequent T-cell expansion is higher when chemotherapy is administered before surgery and a tumor is in situ, providing a rationale for the administration of chemoimmunotherapy in the neoadjuvant setting [31]. The KEYNOTE-173 and I-SPY2 trials investigated the addition of pembrolizumab to standard neoadjuvant chemotherapy for patients with high-risk, early-stage TNBC, with both showing an increase in pCR with the addition of the ICI [10, 32, 33]. The phase II GeparNUEVO study investigated the addition of durvalumab to neoadjuvant chemotherapy. This trial was unique because it included a window-of-opportunity arm in which patients received 2 weeks of durvalumab before starting chemotherapy [34]. Although this study showed only a modest, non-significant numerical improvement in pCR, a significant improvement in long-term survival outcomes was observed, particularly for the subset of patients treated with durvalumab before chemotherapy [34]. The phase III IMpassion031 study assessed the combination of neoadjuvant chemotherapy with or without atezolizumab. pCR was documented in 58% of patients in the ICI arm versus 41% in the chemotherapy arm (difference rate 17%; 95% CI 6–27, $p = 0.004$), and, as mentioned above, the PD-L1 population had a higher rate of pCR (69 vs. 49%; 95% CI 4–35, $p = 0.21$) [20]. These data raise a question about the value of pCR as a surrogate endpoint in neoadjuvant immunotherapy studies. The early findings of the KEYNOTE-173 and I-SPY2 trials were eventually supported and confirmed in KEYNOTE-522, leading to the approval of neoadjuvant chemoimmunotherapy [12]. The early-stage TNBC trials assessing the role of ICIs are summarized in Table 1.

In KEYNOTE-522, all patients randomized to pembrolizumab received neoadjuvant and post-neoadjuvant ICIs, irrespective of treatment response [12]. For patients who achieved a pCR, 3-year survival was 94.4 and 92.5% in the ICI and placebo arms, respectively, with an absolute difference of 1.9%. In contrast, patients with residual disease after neoadjuvant therapy appeared to derive a larger benefit from immunotherapy, with an EFS of 67.4 versus 56.8% (absolute difference 10.6%). In the mentioned IMpassion031 study, patients who were randomized to atezolizumab in the neoadjuvant setting received an additional 11 doses of this medication after surgery; however, the study was not powered to

assess EFS and OS data and longer follow-up is needed [20]. Additionally, the phase III NSABP B-59/GeparDuoze study is also assessing the role of atezolizumab in combination with chemotherapy in the neoadjuvant setting followed by post-neoadjuvant atezolizumab [35]. However, the value of adjuvant ICIs in the setting of pCR remains unclear, as in the mentioned studies, patients are receiving ICIs in the pre- and postoperative setting. The phase III, open-label OptimICE pCR study will randomize patients with early TNBC with pCR after neoadjuvant chemoimmunotherapy to 27 weeks of adjuvant pembrolizumab versus observation, and will likely answer the question about the role of post-neoadjuvant ICIs in these patients.

The role of adjuvant immunotherapy in the absence of neoadjuvant exposure remains unclear. IMpassion030 (NCT03498716) is an ongoing phase III study randomizing patients to a standard anthracycline and taxane-containing adjuvant chemotherapy regimen with or without atezolizumab [36]. Other studies are assessing the role of ICIs for patients with residual disease after neoadjuvant chemotherapy. The SWOG S1418/NRG BR006 study (NCT02954874) is assessing the value of pembrolizumab versus observation in patients with 1 cm or more node-positive residual disease after neoadjuvant chemotherapy [37]. Similarly, the A-BRAVE study (NCT02926196) is assessing the role of avelumab for patients with high-risk early-stage TNBC in the adjuvant or post-neoadjuvant settings [38]. These studies will not answer the question about continuation of ICIs in high-risk patients, but instead are assessing the role of ICIs in patients with high-risk disease treated with neoadjuvant chemotherapy without ICIs.

2.3 Optimal Chemotherapy

Each chemotherapy agent may have differential effects on the tumor and tumor microenvironment (TME), and it is possible that the choise of chemotherapy can impact tumor responses to ICIs. In KEYNOTE-522, a platinum, anthracycline, taxane-containing regimen was used. This treatment is associated with higher rates of short-term toxicities and risk for rare but serious long-term adverse effects such as cardiotoxicity and secondary myelodysplastic syndrome/acute myeloid leukemia (MDS/AML) [12]. The optimal chemotherapy combination for ICIs has not been established. TONIC was a study in which 67 patients with metastatic TNBC received 2 weeks of induction with different chemotherapy agents or radiation, followed by the ICI nivolumab [39]. Doxorubicin showed the most significant activity (objective response rate [ORR] 35%), followed by cisplatin (ORR 23%) [39]. Correlative studies revealed upregulation of PD-L1 and other T-cell cytotoxicity markers with doxorubicin and cisplatin. However, the TONIC study was conducted in patients with metastatic disease, and differences in the TME and PD-L1 expression between primary and metastatic tumors have been previously described [27, 28]. Therefore, we have to be careful about extrapolating the results of the TONIC data to patients with early-stage disease.

One frequent clinical question is if there is a requirement for anthracyclines [14]. In the NeoTRIP study (NCT02620280), 280 patients with early TNBC were randomized to receive an anthracycline-free chemotherapy regimen in the neoadjuvant setting, with or without the ICI atezolizumab; patients then underwent surgery, followed by four cycles of a post-neoadjuvant anthracycline-containing regimen without ICIs [19]. No statistically significant difference in pCR was observed between the groups, and a possible explanation could be an imbalance in TILs [19, 23]. The EFS data (primary endpoint) from NeoTRIP have not been published and may help elucidate whether an anthracycline is needed in this setting.

The addition of platinum to neoadjuvant chemotherapy is also a controversial topic and has been studied in multiple trials with conflicting results; however, previous studies did not assess the role of platinum in patients receiving chemoimmunotherapy. Based on the chemotherapy backbone utilized in the KEYNOTE-522 study, carboplatin and anthracyclines are frequently incorporated into the neoadjuvant regimen for patients with stage II–III TNBC who receive neoadjuvant chemoimmunotherapy.

2.4 Post-neoadjuvant Therapies for Patients with Residual Disease After Chemoimmunotherapy

2.4.1 BRCA Wild-Type

Prior to the approval of the KEYNOTE-522 regimen, patients with residual disease after neoadjuvant chemotherapy were treated with adjuvant capecitabine monotherapy per the CREATE-X study [11]. In KEYNOTE-522, patients in the investigational arm continued pembrolizumab after surgery irrespective of response to therapy, and could not receive additional agents. A common clinical question that arises is whether pembrolizumab should be combined with capecitabine in this setting. Safety data for the combination are available in the metastatic setting [40] and therefore many have considered this in clinical practice; however, additional data are needed to assess the efficacy/potential synergistic effect of the combination of capecitabine and pembrolizumab in patients with early-stage disease.

2.4.2 BRCA Mutation Carriers

The poly(adenosine diphosphate–ribose) polymerase (PARP) inhibitors olaparib and talazoparib have been

Table 1 Studies assessing the role of immunotherapy in early-stage breast cancer

Trial	Regimen	Setting	Sample size (*n*)	Patient population	Endpoints	Results
GeparNUEVO (NCT02685059, phase II) [34]	Window-of-opportunity durvalumab for 2 weeks for the subgroup of patients in the study. Patients were then randomized to Nab-paclitaxel 2 EC ± durvalumab	Neoadjuvant	174	T1b-T4a TNBC	Primary: pCR Secondary: iDFS, DRFS, OS	pCR rate was 53.4 vs. 44.2% favoring the ICI; however, the pCR effect was only seen in the window cohort. 3-year iDFS was 84.9 vs. 76.9% ($p = 0.00559$), DDFS was 91.4 vs. 79.5% ($p = 0.0148$), and OS was 95.1 vs. 83.1% ($p = 0.0076$)
ISPY2 (NCT01042379, phase II) [32]	Paclitaxel ± pembrolizumab, followed by AC	Neoadjuvant	250 (the treatment arm included 69 patients: 40 HR+ and 29 TNBC)	Stage II and III: primary tumors > 2.5 cm	Primary: pCR Secondary: RCB, DRFS	Improvement in pCR with ICIs: ITT 17 vs. 44%, HR+ 13 vs. 30%, TNBC 22 vs. 60%. EFS was 93% at 3 years for those who achieved a pCR
IMpassion031 (NCT3197935, phase III) [20]	Nab-paclitaxel AC ± atezolizumab, followed by adjuvant atezolizumab/placebo	Neoadjuvant and post-neoadjuvant	333	Stage II and III TNBC	Primary: pCR Secondary: EFS, DFS, OS	pCR was 58 vs. 41% favoring ICIs
KEYNOTE522 (NCT03036488, phase III) [12]	Neoadjuvant carboplatin + paclitaxel ± pembrolizumab followed anthracycline + cyclophosphamide ± pembrolizumab. Followed by adjuvant pembrolizumab	Neoadjuvant and post-neoadjuvant	1174	TNBC: tumors larger than >2 cm or with lymph node involvement	Primary: pCR and EFS Secondary: outcomes for the PD-L1+ population	7.5% absolute benefit on pCR favoring the ICI (63 vs. 56%). 3-year EFS 84.5 vs. 76.8% ($p = 0.00031$). Trend towards OS benefit
NeoTRIP (NCT02620280, Phase III) [19]	Carboplatin + nab-paclitaxel ± atezolizumab, followed by an anthracycline-containing regimen after surgery	Neoadjuvant	280	TNBC: tumors larger than > 5 cm or with lymph node involvement	Primary: EFS Secondary: pCR	EFS data not available. pCR was 48.6 vs. 44.4% (HR 0.48)

AC doxorubicin, cyclophosphamide, *DRFS* distant recurrence-free survival, *EC* epirubicin + cyclophosphamide, *EFS* event-free survival, *HR* hazard ratio, *HR+* hormone receptor-positive, *ICI* immune checkpoint inhibitor, *iDFS* invasive disease-free survival, *ITT* intention to treat, *OS* overall survival, *PD-L1* programmed cell death ligand 1, *pCR* pathologic complete response, *RCB* residual cancer burden, *TNBC* triple-negative breast cancer

approved for the treatment of patients with metastatic HER2-negative breast cancer who carry a germline *BRCA1/2* pathogenic or likely pathogenic mutation—around 10% of breast cancer patients [41, 42]. In March 2022, olaparib was approved by the FDA for the adjuvant treatment of patients with HER2-negative high-risk disease who received prior chemotherapy and who had a pathogenic germline BRCA mutation, based on the findings of the OlympiA study [43, 44], a phase III study in which 1836 patients with pathogenic germline BRCA mutations and HER2-negative early breast cancer were randomized to receive 1 year of adjuvant olaparib or placebo [43]. All patients received at least six cycles of chemotherapy, although the inclusion criteria varied for patients treated in the neoadjuvant or adjuvant setting, and according to HR status. The neoadjuvant group included patients with TNBC who did not achieve a pCR [43, 45]. The adjuvant group included patients with TNBC with tumor size larger than 2 cm and/or with any nodal involvement [43]. Overall, 80% of patients in OlympiA had TNBC disease, 72% had a *BRCA*1 mutation, around half received neoadjuvant therapy, 93% received anthracycline and taxane-containing regimens, and 26% also received a platinum compound [44]. With a 3.5-year median follow-up, OlympiA showed a statistically significant improvement in invasive DFS (iDFS), with a 4-year iDFS of 82.7 versus 75.4% with olaparib and placebo, respectively. And with a 4-year OS of 89.8 versus 86.4% favoring olaparib over placebo (stratified HR 0.68, 98.5% CI 0.47–0.97, $p = 0.009$) [44]. When specifically assessing the TNBC population, the iDFS (HR 0.620, 95% CI 0.48–0.78) and OS (HR 0.64, 95% CI 0.45–0.88) was significantly improved. The AE profile was similar to what has been reported in the metastatic setting, with nausea, vomiting, fatigue, and anemia occurring in over 20% of patients in the olaparib arm [43]. The most common grade > 3 toxicities were anemia (8.7%), neutropenia (4.8%), leukopenia (3%), and fatigue (1.8%) [46]. Treatment-related MDS or AML were not reported in higher rates in the PARP inhibitor arm, however longer follow up is needed [47]. The treatment was overall well tolerated and there were no significant differences in quality-of-life outcomes [48].

Prior to OlympiA, patients with germline *BRCA* (pathogenic or likely pathogenic) mutations were treated similarly to those without pathologic mutations, as it pertains to systemic treatment [11, 12]. The approval of olaparib in this setting raised the question of treatment combinations. The combination of PARP inhibitors with several chemotherapy agents (has not been studied with capecitabine) has been challenging due to hematologic toxicities [49]. In contrast, the TOPACIO and MEDIOLA trials showed that the combination of PARP inhibitors with ICIs is well tolerated, although these were single-arm trials, and it remains unclear if the combination of a PARP inhibitor and an ICI is more effective than a PARP inhibitor alone [50, 51]. The ETCTN study (NCT02849496) randomized patients to olaparib with or without atezolizumab and has now completed accrual. This will be the first randomized trial comparing PARP inhibitor monotherapy with a PARP inhibitor in combination with an ICI. There is a strong rationale for this combination as PARP inhibitors appear to increase PD-L1 expression and modify the TME towards a more immunogenic state, having a synergistic effect with ICIs [52, 53]. Again, data are not available for this promising combination for patients with early-stage disease and this is another important question that needs to be answered in order to continue to adapt therapies for our patients.

2.5 Current Management of Early-Stage TNBC

In summary, for patients with early-stage TNBC with tumors 2 cm or larger and/or with lymph node involvement, neoadjuvant therapy is recommended; the KEYNOTE-522 regimen is the current standard of care. Multiple questions remain about the optimal post-neoadjuvant therapy for those who do not achieve a pCR, and about the role of pembrolizumab in those who do. Another important question is whether tumors with low estrogen and progesterone expression (1–10%) should be treated as TNBC, as these tumors appear to have a more aggressive behavior and limited responses to endocrine therapy [54]. Polychemotherapy remains recommended for many patients with small and node-negative TNBC (< 2 cm), but the potential benefit should be assessed on a case-by-case basis. Figure 1 summarizes our current approach to the treatment of patients with early-stage disease.

3 Hormone Receptor (HR)-Positive Breast Cancer

HR-positive, HER2-negative breast cancer is characterized by the expression of estrogen and/or progesterone receptors and represents the most common breast cancer subtype, accounting for around 60–70% of all breast cancers [55]. Most patients with early HR-positive breast cancer receive adjuvant endocrine treatment, which approximately halves the risk of recurrence [56]. Premenopausal patients are often treated with tamoxifen, while postmenopausal patients are often treated with aromatase inhibitors, although tamoxifen is also a treatment option for postmenopausal patients. The role of ovarian function suppression (OFS) in combination with endocrine for the treatment of premenopausal patients was studied in the SOFT and TEXT trials [57]. In SOFT, 2672 patients were randomized to receive tamoxifen, tamoxifen + OFS, or exemestane + OFS. In TEXT, 3066 patients were randomized to tamoxifen + OFS or exemestane + OFS. A combined analysis of these studies, with a median

follow-up of 12 and 13 years, respectively, showed the distant metastases DFS was 88.4% with OFS and exemestane versus 86.6% with tamoxifen and OFS ($p = 0.3$) [58]. There was no difference in OS between the groups, and notably, lower-risk patients, who were not treated with chemotherapy, had a 12-year survival rate of 95%, irrespective of the endocrine therapy selection [58]. In summary, premenopausal patients with low-risk disease are often treated with tamoxifen, while those with high-risk disease may benefit from OFS and an aromatase inhibitor.

The implementation of genomic assays has allowed us to determine which patients with early-stage disease are likely to derive a benefit from the addition of chemotherapy to endocrine therapy. However, other factors such as menopausal status, tumor size, and lymph node burden contribute to the risk of relapse and the potential benefit of chemotherapy [59–61]. Neoadjuvant chemotherapy is not standard in this patient population as the rates of pCR are lower, and correlation with long-term outcomes is weaker compared with other subtypes. Post-neoadjuvant therapy, except for olaparib, is not typically adapted based on the response to neoadjuvant chemotherapy. Notably, the mentioned CREATE-X study included 304 (68%) patients with HR-positive breast cancer [11]; however, these patients did not have a statistically significant benefit with capecitabine treatment. The 5-year PFS was 76.4% with capecitabine and 73.4% in the control group (HR 0.81, 95% CI 0.55–1.17) and OS was 93.4% versus 90% (HR 0.73, 95% CI 0.38–1.40) [11]. Based on these findings, capecitabine is not routinely used in the adjuvant setting for HR-positive breast cancer. The current standard of care for patients with high genomic-risk breast cancer includes adjuvant multichemotherapy regimens; for selected patients, neoadjuvant chemotherapy can be considered, with all patients then going on to receive adjuvant endocrine therapy. Patients with high-risk clinicopathologic features are at high risk of recurrence, warranting the need to assess whether drugs approved for metastatic HR-positive breast cancer can improve outcomes for patients with early-stage disease.

3.1 Cyclin-Dependent Kinase 4 and 6 (CDK 4/6) Inhibitors

Cyclin-dependent kinase (CDK) 4/6 inhibitors have changed the treatment paradigm of advanced HR-positive breast cancer. Three CDK4/6 inhibitors have been approved for use in the metastatic setting in combination with endocrine treatment, i.e. palbociclib, ribociclib, and abemaciclib [62–64]. These drugs demonstrated very similar PFS outcomes in patients with untreated or previously treated advanced disease. In recent years, the use of CDK 4/6 inhibitors has been also studied in patients with early-stage disease.

Two studies have assessed the role of adjuvant palbociclib in early-stage breast cancer. PALLAS was a global, multicenter, phase III study in which 5760 patients with stage II–III HR-positive, HER2-negative breast cancer were randomized to receive 2 years of adjuvant palbociclib plus ongoing endocrine therapy versus endocrine therapy alone, after the completion of standard local and systemic therapies [65]. Patients received the standard dose of palbociclib (125 mg daily, 3 weeks on and 1 week off) and started treatment within 12 months after definitive surgery [65]. About 82% of patients had stage IIB–III cancer, 82% had received chemotherapy, 13% had node-negative disease, 67% had grade 1–2 disease, and about half of the patients were premenopausal [66]. At a median follow-up of 31 months, there was no difference between the groups in iDFS, invasive breast cancer-free survival, distant RFS (DRFS), or OS [66]. In the PENELOPE-B trial, eligible patients had high-risk residual disease after receiving taxane-containing neoadjuvant chemotherapy, defined as a CPS-EG score ≥ 3, or 2 if lymph nodes were involved after surgery (ypN+) [67]. A total of 1250 patients were randomized to receive 1 year of palbociclib versus placebo, both in combination with ongoing adjuvant endocrine therapy [67]. In this study, at a median follow-up of 42.8 months, palbociclib did not improve iDFS compared with placebo [67]. Based on these findings, adjuvant palbociclib is not currently recommended in the early-stage breast cancer setting. The ongoing NATALEE (NCT03701334) study has a similar design as

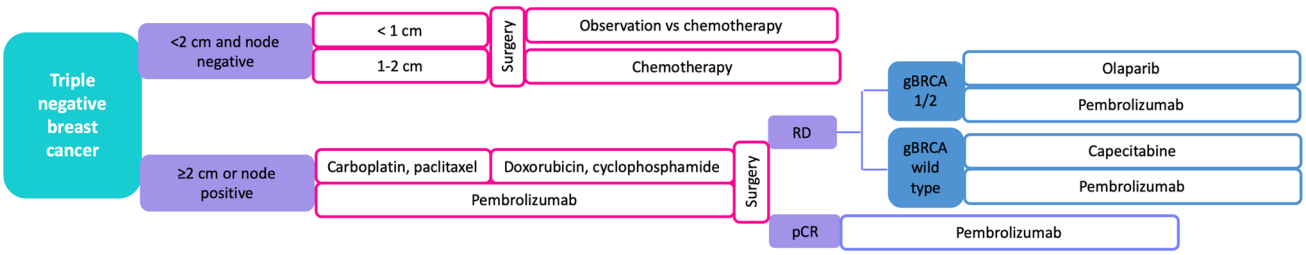

Fig. 1 Proposed treatment algorithm for early-stage triple-negative breast cancer (Data for the combination of olaparib and pembrolizumab and capecitabine and pembrolizumab are not available in this setting. Radiation therapy should be considered on a case-by-case basis). *RD* residual disease, *pCR* pathologic complete response

PALLAS but is using 3 years of reduced-dose ribociclib, with results pending [68].

Abemaciclib has also been studied in the early setting in the phase III monarchE trial. The monarchE study design resembled PALLAS, however patient selection was restricted to patients with four or more lymph nodes involved or one to three lymph nodes with at least one of the following high-risk features: grade 3 (G3) disease, tumor size ≥ 5 cm, or Ki-67 ≥ 20% [69]. A total of 5637 patients were randomized to adjuvant abemaciclib (150 mg twice daily for 2 years) with endocrine therapy versus endocrine therapy alone [69]. With a median follow-up of 27.1 months, the iDFS with abemaciclib was 88.8 versus 83.4% with endocrine therapy alone (HR 0.696, 95% CI 0.58–0.82, $p = 0.0001$) [70]. DRFS was 90.3 versus 86.1%, again favoring the abemaciclib group (HR 0.68, 95% CI 0.57–0.82, $p = 0.0001$) [70]. The most common toxicities reported in the abemaciclib arm were diarrhea (82%, 8% G3), neutropenia (45%, 18% G3), fatigue (38%, 2.8% G3), leukopenia (37%, 11% G3), abdominal pain (34%, 1.3% G3), nausea (28%) and anemia (23%, 2% G3). Venous thromboembolism was reported in 2.3% of the abemaciclib arm and 0.5% of the endocrine therapy-alone arm. These findings led to the FDA approval of adjuvant abemaciclib for patients with high-risk (G3, T3 or more, N2 or higher), early-stage HR-positive breast cancer and a Ki-67 ≥ 20% in October of 2021. Data from cohort 1 (four or more lymph nodes, one to three lymph nodes plus G3 or > 5 cm) showed an iDFS of 88.6 versus 82.9% favoring abemaciclib without the Ki-67 restriction (HR 0.680, 95% CI 0.57–0.80, $p = 0.0001$) [71]. Given that patients seem to derive a benefit from abemaciclib irrespective of Ki67 results, American Society of Clinical Oncology (ASCO) and National Comprehensive Cancer Network (NCCN) guidelines recommend consideration for abemaciclib in all high-risk patients who would have qualified for monarchE, not only in the subset with Ki-67 ≥ 20% [72]. Notably, abemaciclib was approved in combination with tamoxifen or aromatase inhibitors in the adjuvant setting, and is only approved in combination with aromatase inhibitors in the metastatic setting. The combination of abemaciclib and tamoxifen was associated with a 4.3% rate of venous thromboembolism versus 1.8% with abemaciclib and aromatase inhibitors; therefore, this combination should be used with caution [73].

As the PFS benefits in the advanced setting for CDK4/6 inhibitors have been consistent, it remains unclear why the studies in the early setting are conflicting. One possible explanation has been that patients in monarchE had higher-risk disease. However, a recent analysis from PALLAS showed that even patients with high-risk disease did not benefit from palbociclib, and patients in PENELOPE-B, with non-pCR, could be considered the highest risk of all patients enrolled [66]. The duration of treatment, i.e. 2 years, was the same in PALLAS and monarchE; however, abemaciclib is administered continuously and palbociclib is administered 3 weeks on and 1 week off. It has been proposed that the difference in dosing schedules could have led to differences in outcomes, as there is some concern that intermittent dosing may not be as tumor-suppressive in the curative setting. In addition, pharmacodynamic assays show a different kinase specificity with abemaciclib compared with palbociclib and ribociclib, which might account for the different safety profile and the efficacy reported [74, 75]. The discontinuation rate was higher in PALLAS when compared with monarchE, however analyses looking at outcomes in PALLAS based on dose intensity suggest this may not be the case [66]. Additional analyses will likely help us determine which patients are at higher risk for recurrence and to predict who would derive a benefit from the addition of a CDK4/6 inhibitor. The characteristics of the studies assessing the role of CDK4/6 inhibitors in early-stage breast cancer are summarized in Table 2.

3.2 Current Management of Early-Stage HR-Positive Breast Cancer

Patients with high-risk HR-positive breast cancer who do not carry a BRCA mutation are often treated with local therapies, followed by chemotherapy (if indicated), with radiation therapy added on a case-by-case basis. Figure 2 summarizes our current approach to the treatment of patients with early-stage disease.

For patients with germline *BRCA1/2* pathogenic or likely pathogenic variant associated with high-risk HR-positive early breast cancer, the PARP inhibitor olaparib is now approved based on the findings of the OlympiA study [43], discussed above. Patients with HR-positive disease were included in the neoadjuvant group if they did not achieve a pCR and had a CPS-EG score ≥ 3, and in the adjuvant group if they had four or more lymph nodes involved [43, 45]. There was an improvement in OS and iDFS in the intention-to-treat population. When assessing the HR-positive population only, the benefit appeared to be smaller (iDFS: HR 0.68, 95% CI 0.40–1.3; OS: 0.69, 95% CI 0.39–1.18), which could be attributed to the smaller sample size [43, 45]. This approval raised the question about treatment selection for patients eligible to receive both olaparib and abemaciclib. The HR-positive group represents only 18% of the study population of OlympiA; this group was added as an amendment to the study while the trial was ongoing. Given the small sample size, the CIs surrounding efficacy estimates in this cohort are much wider. Given the significant DFS and OS benefit seen within OlympiA, utilizing it in patients with HR-positive breast cancer is reasonable. Additionally, given the lack of OS benefit at this time from monarchE, preference for utilization of olaparib should be made over

Table 2 Studies assessing the role of CDK 4/6 inhibitors in hormone receptor-positive early-stage breast cancer

	PALLAS (NCT02513394, phase III) [65]	PENELOPE-B (NCT01864746, phase III) [67]	MonarchE (NCT03155997, phase III) [69]	NATALEE (NCT03701334, phase III)
Treatment (drug, dose, duration)	Adjuvant palbociclib, 125 mg daily, 21 days on, 7 days off, for 2 years with ET vs. ET alone	Post-neoadjuvant palbociclib, 125 mg daily, 21 days on, 7 days off, for 1 year with ET vs. ET alone	Adjuvant abemaciclib, 150 mg twice daily continuous for 2 with ET vs. ET alone	Adjuvant ribociclib, 400 mg daily 21 days on 7 days off for 3 years with ET vs. ET alone
Patient population	Stage II–III	Residual disease after receiving a taxane-containing neoadjuvant chemotherapy regimen and at risk for relapse, which was defined as a CPS-EG grade > 3, or 2 if lymph nodes were involved after surgery (ypN+)	Four or more lymph nodes involved or patients with one to three lymph nodes with at least one of the following: grade 3 disease, tumor size ≥ 5 cm, Ki 67 > 20%	Stage II–III
Sample size	5760	1250	5637	5101
iDSF	No difference in iDFS led to study discontinuation	No difference in iDFS	iDFS with abemaciclib was 88.8 vs. 83.4% with endocrine therapy alone (HR 0.696, 95% CI 0.58–0.82, $p = 0.0001$)	Not available, trial ongoing
OS	Not available	Not available	Longer follow-up is needed	Not available, trial ongoing

CDK cyclin-dependent kinase, *CI* confidence interval, *ET* endocrine therapy, *HR* hazard ratio, *iDFS* invasive disease-free survival, *OS* overall survival

abemaciclib for most high-risk germline BRCA patients with HR-positive disease. Alternatively, sequencing of therapies might be an option; very high-risk patients could potentially receive 1 year of olaparib followed by 2 years of abemaciclib, however there are no data on the efficacy or safety of treatment sequencing with these prolonged therapies for patients with potentially curable disease.

4 Human Epidermal Growth Factor Receptor 2 (HER2)-Positive Breast Cancer

HER2-positive breast cancer accounts for about 20% of all breast cancers, and approximately half of these tumors also express HRs [76, 77]. HER2-positive breast cancer was historically associated with a worse prognosis than other breast cancer subtypes; however, HER2-directed therapies have dramatically improved outcomes for these patients [46, 78]. Several types of HER2-targeted therapies have been studied and approved in early-stage HER2-positive breast cancer, including the mAbs trastuzumab and pertuzumab, the tyrosine kinase inhibitor (TKI) neratinib, and the antibody-drug conjugate (ADC) trastuzumab emtansine (T-DM1).

Patients with tumors ≤ 2 cm and without lymph node involvement often undergo surgery and then receive adjuvant therapy with paclitaxel and trastuzumab (TH) for 12 weeks followed by trastuzumab to complete 1 year of treatment, based on the APT trial [77]. TH has been associated with excellent long-term outcomes, with a 7-year iDFS of 93% and OS of 95% [79]. The role of T-DM1 was studied in the ATEMPT trial, in which patients with stage I, HER2-positive breast cancer were randomized to receive 1 year of T-DM1 versus TH. The study found that 1 year of T-DM1 was associated with a 3-year iDFS of 97.8%, but found there was no significant difference in clinically relevant toxicities between the arms [80]. The ongoing ATEMPT 2.0 trial (NCT04893109) is randomizing patients to six cycles of T-DM1 followed by trastuzumab to complete 1 year of treatment versus the mentioned TH regimen, and will help determine if further shortening the T-DM1 duration allows maintenance of excellent outcomes while minimizing toxicities. The ongoing ADEPT trial (NCT04569747) is assessing the role of an adjuvant chemotherapy-free regimen containing trastuzumab, pertuzumab (HP) and endocrine therapy for patients with stage I HR- and HER2-positive breast cancer.

Patients with larger or node-positive tumors often receive neoadjuvant therapy, with combination chemotherapy and trastuzumab with or without pertuzumab [47]. Administration of systemic therapy prior to surgery is critical in patients with stage 2–3 HER2-positive breast cancer since pCR has shown to be a good prognostic marker in this tumor subtype [10], and perhaps more importantly because post-neoadjuvant therapy can now be adapted based on the patient's

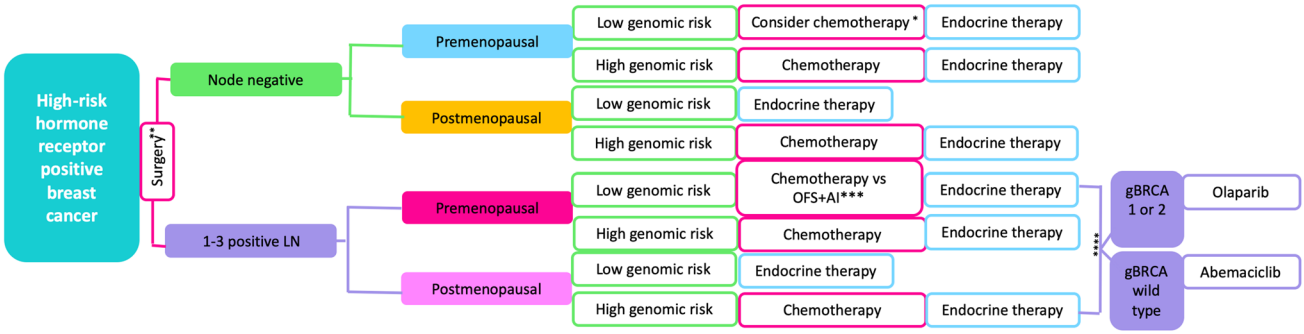

Fig. 2 Proposed treatment algorithm for early-stage hormone receptor-positive breast cancer. Radiation therapy should be considered on a case-by-case basis. *The potential benefit of chemotherapy is based on genomic assays. High risk can be defined based on tumor size and nodal status. For premenopausal patients with intermediate genomic risk, assessing clinical risk can help determine the potential benefit of chemotherapy, however for those with low genomic risk there is no benefit of chemotherapy. **Neoadjuvant therapy can be considered on a case-by-case basis. ***For premenopausal patients with lymph node involvement and low genomic risk, ovarian suppression and aromatase inhibitor could be considered as an alternative to chemotherapy. ****Olaparib or abemaciclib should be initiated with endocrine therapy. Olaparib is US FDA-approved for patients with pathogenic BRCA 1 or 2 mutations who have high-risk disease and have been treated with chemotherapy. Abemaciclib is FDA-approved for patients with node-positive disease and Ki67 ≥ 20%, while the NCCN and ASCO guidelines recommend use in patients with one to three positive lymph nodes and either G3 or ≥ 5 cm. *OFS + AI* ovarian function suppression + aromatase inhibitor, *NCCN* National Comprehensive Cancer Network, *ASCO* American Society of Clinical Oncology

response to the initial therapy [6, 46]. A recent update from the APHINITY trial showed that with a median follow-up of 8.4 years, patients with lymph node involvement at baseline benefitted from adding pertuzumab to their therapy [81]. These patients had a 4.9% absolute improvement in iDFS and 1.9% in OS; the benefit was not seen in patients with node-negative disease [81]. There was no difference by HR status [81]. Given that the lymph node status is difficult to assess at baseline, patients with higher-risk tumors often receive chemotherapy in combination with HP [81]. For patients with no clinical evidence of lymph node involvement at baseline and who achieve a pCR to preoperative therapy, trastuzumab monotherapy could be considered in order to complete 1 year of treatment; for those with lymph node involvement at baseline who achieve a pCR, treatment with HP to complete 1 year of treatment should be considered based on APHINITY [81].

Patients with residual disease should receive adjuvant T-DM1 for 14 cycles, based on the findings of the KATHERINE study, which demonstrate that T-DM1 decreased the risk of recurrence by 50% when compared with trastuzumab alone [78]. The iDFS at 3 years was 88.3 versus 77% for adjuvant T-DM1 versus adjuvant trastuzumab, respectively [6]. Eighty percent of patients in this study received trastuzumab-based HER2-targeted therapy. Notably, half of the distant recurrences in the T-DM1 arm occurred in the central nervous system (CNS), which was similar to the trastuzumab arm, suggesting T-DM1 did not impact rates of CNS events [46]. There is an unmet need to optimize therapies to prevent CNS disease, which is often associated with significant morbidity and mortality.

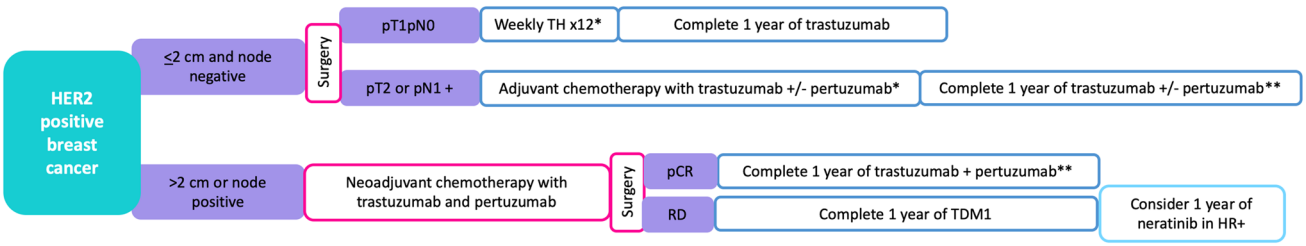

Fig. 3 Proposed treatment algorithm for early-stage HER2-positive breast cancer. Radiation and endocrine therapy should be considered on a case-by-case basis. *For selected patients, 1 year of T-DM1 could be considered as an alternative to TH. **Pertuzumab is often added to neoadjuvant regimens, especially for those with node-positive disease at diagnosis. Patients with lymph node involvement benefit from the addition of adjuvant pertuzumab, however this antibody may be omitted for those without lymph node involvement at diagnosis. *HER2* human epidermal growth factor receptor 2, *HR+* hormone receptor-positive, *RD* residual disease, *pCR* pathologic complete response, *TH* paclitaxel and trastuzumab, *T-DM1* trastuzumab emtansine, *HER2* human epidermal growth factor receptor 2

Table 3 Studies assessing the role of novel adjuvant therapies in HER2-positive breast cancer

	Pertuzumab	T-DM1	Neratinib
Trial	APHINITY (NCT01358877, phase III)	KATHERINE (NCT01772472, phase III)	ExteNET (NCT00878709, phase III)
Treatment	Adjuvant chemotherapy + trastuzumab ± pertuzumab	Adjuvant T-DM1 vs. trastuzumab for patients with residual disease after neoadjuvant therapy with a taxane and trastuzumab (± anthracycline)	Extended adjuvant treatment with neratinib for 1 year vs. placebo after trastuzumab-based therapy
Sample size	4805	1468	2840
Patient characteristics	35% no LN involvement, 64% HR+ disease, 77% received anthracycline-containing adjuvant therapy	72% had HR+ disease, 75% had operable breast cancer, 77% received anthracycline-containing neoadjuvant therapy. Neoadjuvant HER2-directed therapy: 80% trastuzumab alone, 18% trastuzumab + pertuzumab, 1% trastuzumab + another HER2-targeted agent	24% had no LN involvement, 57% had HR+ disease. All patients received trastuzumab, none received pertuzumab or T-DM1
Results	LN+: iDFS was 86.1% and 81.2% favoring pertuzumab (HR 0.72, 95% CI 0.60–0.87); OS was 89.2 vs. 91.1% (HR 0.80, 95% CI 0.63–1.00) LN−: no benefit Findings irrespective of hormone receptor status	iDFS was 12.2% in T-DM1 vs. 22.2% in the trastuzumab arm. 3-year DFS was 88.3 vs. 77% favoring T-DM1 (HR 0.50, 95% CI 0.39–0.64, $p = 0.001$)	The absolute benefit in patients with HR+ disease who received a TKI within 1 year of completion of trastuzumab was 5.1% (HR 0.58, 95% CI 0.41–0.82) and 1.3 (HR 0.74, 95% CI 0.29–1.84) for those with HR+ disease who received a TKI after 1 year

CI confidence interval, *DFS* disease-free survival, *HER2* human epidermal growth factor receptor 2, *HR* hazard ratio, *HR+* hormone receptor-positive, *iDFS* invasive disease-free survival, *LN* lymph node, *OS* overall survival, *T-DM1* trastuzumab emtansine, *TKI* tyrosine kinase inhibitor

Neratinib, a pan-HER2 irreversible TKI, is approved as extended adjuvant therapy for patients with HER2-positive disease following the completion of trastuzumab-based therapy based on the results of the ExteNET trial. ExteNET was a phase III study in which 2840 patients were randomized to receive 1 year of neratinib versus placebo after the completion of trastuzumab-based treatment [82]. With a median follow up of 5 years, the absolute iDFS benefit was 5.1% for patients who started the TKI within 1 year of trastuzumab therapy; the benefit was modest for those who started within 1–2 years [82]. There was also a numerical benefit in OS (91.5 vs. 89.4%; HR 0.79, 95% CI 0.55–1.13, $p = 0.2$), and fewer CNS recurrences favoring neratinib [82]. The benefit of neratinib was primarily seen in patients with HR-positive disease and those who did not achieve a pCR [11]. The most common toxicity was diarrhea, which was reported by 94% of patients, including 39% with grade 3–4; therefore, aggressive antidiarrheal strategies are recommended to improve the tolerability of this medication [82, 83]. The fact that patients did not receive drugs that are currently considered standard of care, such as pertuzumab or T-DM1, is an important limitation of ExteNET. It is unclear if the benefit of adding neratinib would still be present for patients treated with these agents. Ongoing studies are assessing the role of novel TKIs and ADCs in the post-neoadjuvant setting for patients with residual disease after neoadjuvant therapy. CompassHER-RD (NCT04457596) is assessing whether adding tucatinib, a highly selective HER2 TKI, to T-DM1 in the post-neoadjuvant setting could help prevent recurrence. There is also interest to see if tucatinib can help prevent CNS recurrences, given data in the metastatic setting suggesting robust activity in the CNS [84]. Destiny-Breast 05 (NCT04622319) is assessing if the ADC trastuzumab deruxtecan (T-DXd) is superior to T-DM1 in the adjuvant setting for the prevention of recurrence in patients with residual disease after neoadjuvant therapy. Comparing these two agents in the metastatic setting showed that T-DXd is superior to T-DM1 and replaced the latter as second-line therapy [85]. Figure 3 summarizes our current approach to treating patients with early-stage HER2-positive disease, and Table 3 summarizes the trials studying HER2-targeted agents in the adjuvant setting.

5 Conclusions

The treatment landscape for early-stage breast cancer is evolving rapidly. Multiple novel therapies have been approved over the past few years to treat different breast cancer subtypes, increase the available treatment options, and improve outcomes for breast cancer patients.

As adjuvant treatments become more complex, the risk for long-term toxicities of some of these therapies must be considered to guide clinical decisions. Including financial toxicities of these therapies for both patients and healthcare systems, since as the complexity of treatment increases, the cost has increased significantly, and tailoring patients' treatment in order to prevent physical and financial toxicity while maintaining the best possible patient outcomes, needs to be considered. For example, the use of dual HER2-targeted therapy with trastuzumab and pertuzumab increases the cost of treatment significantly; however, an adaptive approach based on response to treatment and lymph node involvement can spare toxicity and reduce treatment cost [86]. Biomarkers are needed to determine how to select and potentially sequence some of these therapies in order to maximize clinical benefit and limit long-term toxicities for patients with curable disease. Another consideration is that access to cancer treatments is not equal worldwide, mostly due to regulatory and cost barriers (87). These disparities need to be addressed urgently in order to continue to improve outcomes for all patients.

Declarations

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